

Dhruv Patel

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Education

- **Stony Brook University** Aug 2025 – May 2027
Master of Science in Computer Science, GPA: 3.61/4.00
Stony Brook, NY
- **Dhirubhai Ambani Institute of Information and Communication Technology** Nov 2020 – May 2024
B.Tech. (Honours) in ICT, Minor in Computational Science, GPA: 3.45/4.00
Gandhinagar, India

Technical Skills

Languages: C++, Python, C, Java, SQL

Systems: Linux, Multithreading, Socket Programming, Distributed Systems, Concurrency, System Design, Performance Optimization

Computer Science: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Software Engineering, Debugging, Software Testing

Backend: FastAPI, Flask, REST APIs, Node.js

Cloud: Docker, Kubernetes, AWS, Azure, Google Cloud Platform, Git, GitHub

Databases: PostgreSQL, Redis, MongoDB, MySQL

Machine Learning: PyTorch, TensorFlow, FAISS, HuggingFace, scikit-learn

Projects

- **Distributed File Sharing System** *C++, Linux, TCP Sockets*
Implemented a multithreaded peer-to-peer file sharing system in C++ using TCP sockets on Linux, supporting concurrent clients and reliable file transfer.
- **Raft Consensus Algorithm Implementation** *C++, Distributed Systems*
Implemented the Raft consensus algorithm with leader election, log replication, and fault-tolerant recovery, validating correctness through simulated node failures and network partitions.
- **ARGUS: Local AI Site Compliance Platform** *Python, FastAPI, CUDA, PyTorch*
 - Developed a local AI pipeline for construction-site safety by integrating video analysis, retrieval, regulatory grounding, and GPU-accelerated inference into a unified backend.
 - Optimized runtime through CUDA-backed inference, model prewarming, batching, and detector reuse while exposing scalable FastAPI services for real-time analysis.
- **OnboardOps: Repository Onboarding Platform** *FastAPI, WebSockets, Next.js*
Built an event-driven onboarding platform using FastAPI, WebSockets, and an MCP server to automate repository analysis and stream real-time architectural insights to an interactive dashboard.

Experience

- **Research Project** Jan 2024 – Jul 2024
Accelerating Particle-in-Cell Simulations Using Deep Learning (APS GEC 2025)
DA-IICT, India
 - Developed high-performance scientific simulation software using Python and C extensions on Linux, reducing execution time by 45% while maintaining 96–98% predictive accuracy.
 - Optimized memory allocation and numerical computation for large-scale scientific workloads while integrating deep learning into Particle-in-Cell simulations.
 - Integrated deep learning models into traditional Particle-in-Cell simulations, accelerating computation while preserving simulation accuracy.
- **Summer Research Intern** May 2023 – Aug 2023
Software Testing and Test Case Prioritization
DA-IICT, India
 - Implemented and benchmarked multiple test case prioritization algorithms to improve fault detection efficiency across large software test suites.
 - Profiled execution bottlenecks, optimized algorithm performance, and automated evaluation pipelines using Python, improving testing efficiency by approximately 30%.

Leadership

- **Deputy Convenor** Apr 2021 – Apr 2023
Annual Techno-Cultural Festival Committee
DA-IICT, India
 - Led technical teams responsible for infrastructure and execution of one of Gujarat's largest techno-cultural festivals serving 25,000+ attendees.